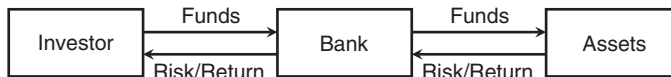


Covered Bonds

Covered bonds are a class of secured bonds issued by financial institutions. The covered bond segment is a large part of the European fixed income universe, and in Denmark exceeds the size of the domestic government bond market. Unlike most other bank bonds, covered bonds tend to have fixed coupon bullet structures to make them more similar to government bonds. This type of bond evolved from mortgage lending, and played an important role in funding large-scale real estate developments, for instance the rebuilding of Copenhagen after a large fire in 1795.

Essentially, a covered bond is a means to reduce the funding cost of banking assets by providing the clients providing the funds with better security than a normal bank debenture or deposit. For a lender to a bank, the funding and risk picture looks like this:



An investor putting money into a bank is exposed to the risks of the bank's assets, as well as additional risks related to the management of the bank. Because the assets of the bank are not directly visible to an investor, some assumptions will be made by the public which may be or not be accurate. If a bank can be set up so that a) investors are well informed about the assets held by the bank, b) these assets are of a high quality, and c) investors have access to these high-quality assets if the bank should fail, then investors should require lower risk premia for their invested money. This is the guiding idea behind the concept of covered bonds which were introduced as a way to source funding for real estate investments at a low cost that reflects the value of having real estate collateral.

Covered bonds are debt instruments issued by banks which are secured by specific collateral of high, and prescribed, quality. Should the issuing bank fail, investors in these bonds gain rights to the assets securing the covered bonds. In addition, covered bonds are excluded from regular insolvency proceedings and do therefore not accelerate in case of default of the issuing bank.

The name 'covered bond' stems from the existence of a cover pool which is a segregated pool of assets on the issuer balance sheet that serves as collateral for the bond. Issuers tend to maintain cover pools that collateralise more than one bond. In

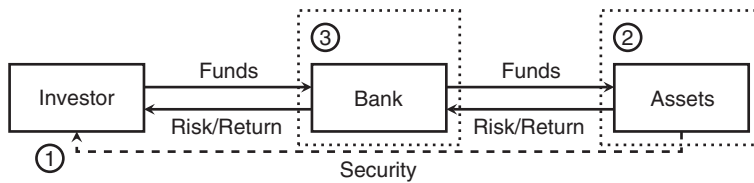


FIGURE 25.1 The essential safety features of a covered bond: Security pledge (1), Cover asset quality (2) and specific bank supervision or specialist bank principle (3).

the Danish market, cover pools are also known as capital centres. The three essential safety features of covered bonds are outlined in Figure 25.1. The particular safety of these bonds arises from the combination of them being secured by collateral, the legally prescribed high quality of that collateral, and a specific supervisory regime. The latter can go as far as business restrictions for covered bond issuing banks, known as the specialist bank principle.

Covered bonds provide banks with funding at levels that are far cheaper than their senior unsecured bonds, sometimes by several hundred basis points. Weaker banks can sometimes issue covered bonds even when there is no market for their unsecured debt. In turn, this allows banks to lend at cheaper levels when the loans qualify as cover assets. Seen as a system, covered bonds channel investor funds to specific borrower types at lower rates than the general banking system would be able to. This is the motivation for governments to create covered bond legislation.

Two basic varieties of covered bond exist. The first, statutory covered bonds, rely on specific legislation and supervision to ensure the integrity and asset quality of the cover pool as well as the structure of the covered bond. Covered bond legislation now exists in most developed economies, and several countries have multiple legal frameworks for covered bonds. In countries where covered bond legislation does, or did, not exist, or where issuers find it hard to comply with such legislation, so-called structured covered bonds have been issued. These bonds replicate the core features of a covered bond through contractual means, similar to asset-backed structures. Note that 'structured' in this context does not refer to the coupon structure of the security but the underlying collateral structure. Both statutory and structured covered bonds have evolved over recent years and adopted credit-enhancement features of ABS. The core difference between covered bonds and ABS, however, remains in place: covered bonds are bank liabilities, and the assets that secure these liabilities remain on the balance sheet of the issuer.

The covered bond market traces its existence to the German Pfandbrief, a class of privileged bond introduced in 1770. The Pfandbrief was initially designed as a way to syndicate mortgage loans. Given that mortgage credit had been in existence in bilateral non-bank lending between private individuals, the Pfandbrief can be seen as an industrialised version of a pre-existing form of mortgage lending. The lower interest rates on bank mortgages that are refinanced with Pfandbriefe incentivised replication of this model. The Great Fire of Copenhagen 1795 required a substantial reconstruction effort and the necessary funding was raised with the first Danish covered bonds.

The Pfandbrief structure has since been applied to public sector loans as well. During the halcyon days of the German Pfandbrief in the first few years after the

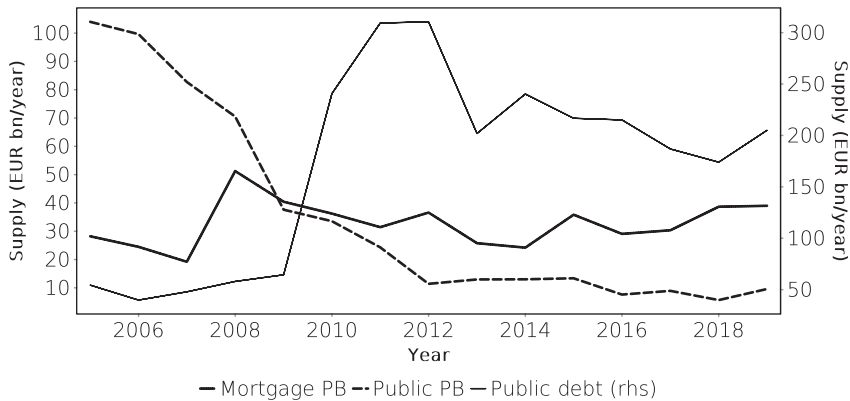


FIGURE 25.2 Annual issuance volumes of mortgage Pfandbriefe, public sector Pfandbriefe, and public sector bonds, excluding German federal government bonds. Source: Data from Bundesbank capital market statistics and German Federal Debt Management Agency.

introduction of the euro, public Pfandbriefe (covered bonds backed by public sector assets) far outstripped the volume of mortgage Pfandbriefe. Two events changed this picture. The first is the expiry of state guarantees ('Gewährträgerhaftung') of liabilities of German state banks (Landesbanken). These were grandfathered in a settlement between Germany's private and public banks that was reached in 2001 and applied mostly to liabilities originated before 18 July 2005 and maturing before the end of 2015. This expiry rendered a large volume of state bank liabilities ineligible as covered bond cover assets. The second was the financial crisis starting in 2007 which reduced the ability of banks in general to intermediate the funding of high-grade public sector debtors through loans. Instead, these public sector borrowers turned to direct capital market financing. While banks continued to purchase such securities, their funding costs no longer gave them a comparative advantage in funding such debtors. The result was a decline in public sector Pfandbrief issuance while public sector bond issuance surged (cf. Figure 25.2). In comparison, mortgage Pfandbrief issuance volumes remained comparatively stable over the years, confirming the importance of this asset class for the German housing market. The Spanish equivalent of the public sector Pfandbrief, *cédulas territoriales*, is now also inactive while the market for mortgage *cédulas hipotecarias* remains vibrant. Some other countries did not develop a public sector covered bond instrument because they already had an existing funding instrument for non-government public sector debt through specialised lenders, such as *Kommunekredit* in Denmark and *MuniFin*¹ in Finland.

Following the introduction of the euro, the importance of competitive bank refinancing channels became more of a focus of legislators and other euro area countries either revived existing legal frameworks or created new ones to provide access

¹The actual name of this institution is *Kuntarahoitus*, often referred to as *Kunta*.

to the covered bond market to their banks. The resulting instruments are collectively known by the generic term ‘covered bond’ because the national designations are generally legally protected. It is illegal in Germany (as well as Austria and Switzerland) to label a French *Obligation Foncière* a ‘Pfandbrief’ even though this bond type has many features in common with a German *Pfandbrief*. Conversely, a German *Pfandbrief* cannot be labelled an *Obligation Foncière* in France.

There is no cross-country, or even EU-wide, standard definition of what exactly constitutes a covered bond. In response to the size constraints of the Danish fixed income market², the European Union has introduced a concept into European legislation that reflects the salient features of most covered bond laws. Generally speaking, the securities satisfying this definition:

- are issued by specialist banks,
- under special supervision by their regulator,
- offer special protection to their holders, and
- are secured by high-quality assets.

Structured covered bonds, by definition, do not satisfy the first two requirements because they rely on contractual means, rather than specific legislation, to provide the last two features.

One of the features that is now generally accepted as necessary in covered bonds is that they should not fall due in case of insolvency of the issuer. Instead, bond holders look to continue receiving the cashflows they originally expect because the security pledged as collateral is expected to generate the necessary cash. This aim requires not only fairly strict asset-liability matching from the issuing bank, but also presents important problems for the legal system under which these bonds are issued. In particular, the applicable bankruptcy code must recognise a situation where insolvency occurs but some creditors remain outside the normal bankruptcy proceedings. This means that the security structure of covered bonds cannot easily be created without close involvement of the sovereign.

Beyond the commonalities lie various differences. Table 25.1 shows examples of differences in such important parameters as maximum permissible loan-to-value (LTV) ratios, lending areas, permissibility of MBS collateral, and so on. The purpose of this table is not to define the various frameworks, but to highlight differences. These differences result in varying degrees of risk and diversification across the different covered bond structures. As will be discussed later, investors need to weigh these differences against other parameters, such as over-collateralisation and maturity profiles, to arrive at a risk assessment.

One driver of an evolution in covered bond structures has been the competition of the ABS market. Because ABS issuing vehicles are capital-light, investors insist on detailed protection mechanisms against risks that could normally be expected to

²The covered bond market in Denmark is much larger than that for government bonds, and therefore covered bonds dominate the positions of domestic asset managers. The resulting levels of issuer concentration are higher than what is normally considered permissible for fund managers.

TABLE 25.1 Simplified comparison of selected features of statutory mortgage covered bond frameworks. In some fields the most relevant value has been selected from multiple options (Source: European Covered Bond Council).

Framework	Max. LTV	Lending area	Substitute	MBS
German Pfandbrief	60%	EEA, Switzerland, US, Canada, Japan, Aus, NZ	20%	×
French Obligation Foncières	80%	EEA, others	15%	•
French Obligation à l'Habitat	80%	France	–	•
Spanish Cédulas Hipotecarias	80%	EU	5%	◦

TABLE 25.2 Outstanding volumes of covered bonds by country at the end of 2018 absolute and per head of the population (Source: Based on European Covered Bond Council, OECD).

Country	Total outstanding (EUR m)	Per capita (EUR)
Australia	65,855	2,677
Austria	58,928	6,717
Canada	107,496	2,949
Denmark	405,991	70,622
Finland	37,257	6,770
France	321,311	4,810
Germany	369,747	4,481
Iceland	3,123	9,231
Ireland	23,319	4,874
Italy	168,936	2,788
Japan	1,000	8
Luxembourg	6,103	10,332
The Netherlands	94,797	5,550
Norway	121,182	23,046
Spain	231,615	4,978
Sweden	217,979	21,808
Switzerland	119,422	14,184
United Kingdom	98,192	1,491

be addressed by a reasonably well-capitalised bank, such as short-term cashflow mismatches, derivatives counterparty risks, etc. Covered bond legislation has added comparable features for cases where the issuing bank is no longer viable, and the protection afforded by its capital is no longer meaningful. The process of managing the cover pool for the benefit of covered bond holders, for instance, is now more regulated than it was a few decades ago.

Aside from the stability of the covered bond product itself, one should be aware that there is strong systemic support for these debt instruments in several countries. Table 25.2 compares outstanding volumes of these bonds to the population. Where covered bonds form a substantial part of the total debt structure in a country, the contagion risk that would arise from the default of any one issuer may make it preferable for other

actors to step in with a support package rather than accept such a default. Such support may take the form of mergers or takeovers between weaker and stronger issuing banks, for instance.

The recent financial crisis has called into question the ability of issuing banks to maintain the cash flow structure required to pay off the covered bonds outstanding at the time they encounter financial difficulties. As a result, some banks have modified their outstanding covered bonds in such a way that the redemption structure can change according to incoming cash flows from the underlying loan pools. In these conditional pass-throughs, the cash flow timing and credit risk passes to investors when the situation of the covered bond issuer no longer permits the assumption of this risk. Another structure that emerged from the crisis are soft bullet covered bonds, structures that can extend maturity under specific circumstances to ensure that redemption cash needs do not lead to an insolvency of the structure. Both conditional pass-throughs and soft bullets should be recognised as forms of credit rating arbitrage. By softening the obligations of the issuer, it is more likely that the issuer can satisfy them, which results in better ratings. For investors, however, very egregious forms of rating arbitrage can create substantial uncertainty about future cash flow timing.

25.1 STATUTORY COVERED BONDS

In a classic, statutory covered bond structure, cover assets are held on the balance sheet of the bond issuer in what is called the cover pool. A special registrar maintains lists of assets held in the cover pool as collateral against the covered bonds and the outstanding bonds secured by these assets. The registrar is responsible for ensuring that the volume of cover assets at all times suffices to cover the outstanding bonds, and that the asset quality is of the required standard. Depending on the jurisdiction, public sector loans and mortgage loans serving as cover assets can be held in the same cover pool or must be in segregated pools. Covered bond issuers are required by law to publish regular updates on the amount and composition of cover assets to allow investors a minimum degree of credit analysis of the pool.

Most jurisdictions permit the limited use of substitute collateral which are assets that are outside the general definition of eligible cover assets. The purpose of substitute collateral is cashflow management and liquidity of substitute assets should generally be very high. For instance, mortgages tend to have monthly amortising structures while covered bonds have annual or semi-annual coupon and bullet structures. The cover pool maintainer may therefore use cash or cash equivalents in the pool to retain cash received from the underlying loans until the next upcoming cash outflow for bond servicing. Over the years, regulators and rating agencies have tightened their scrutiny of substitute collateral.

Cover pools generally contain more cover assets than required to collateralise the outstanding bonds, in a practice called overcollateralisation. The excess collateral is therefore usually called OC. Legislation defines a minimum level of OC, but issuers routinely exceed this legal minimum. The amount of OC, together with the cover pool asset quality and the credit strength of the issuer, largely determine the credit quality of the covered bond. It should be borne in mind that OC is by definition funded

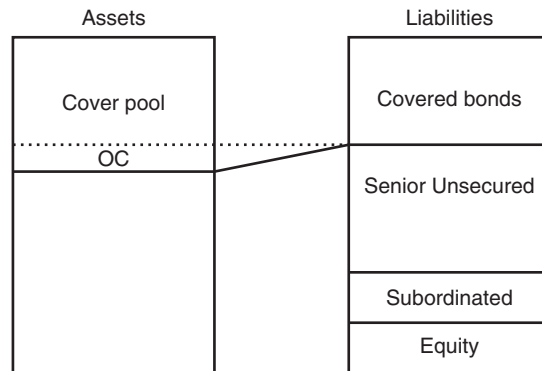


FIGURE 25.3 Balance sheet structure of a covered bond issuer.

by liabilities other than covered bonds, but is pledged to covered bond holders. This asset encumbrance lowers the likely recovery values for unsecured creditors of the issuing bank, including depositors. Some regulators therefore take a rather dim view of covered bonds and severely restrict the volume of covered bonds that can be issued by banks. In other jurisdictions, covered bonds tend to be issued by specialist banks and it is assumed that the other creditors of such banks are aware of the amount of assets that are so encumbered and factor this into their investment decisions. In any case, OC is funded at a cost that usually exceeds the cost of covered bond funding. An issuer must therefore optimise the OC level to be high enough to succeed in placing covered bonds at low interest rates, but not so high as to lead to higher unsecured funding costs due to excessive encumbrance, and higher overall funding cost for the total collateral pool.

Issuers are required to maintain the quality of the cover pool at all times. In practice, this requires the replacement of deteriorated assets with others over time. For instance, an issuer experiencing arrears on a mortgage used in the cover pool must, by law, remove that mortgage from the pool and replace it with a performing loan, under the scrutiny of the cover pool registrar.

The type of loan eligible as a cover asset is restricted by law. Such restrictions take the form of geographic limits for the obligors, as well as loan details. For mortgages, this might be the type of property that secures the mortgage as well as the loan-to-value ratio³ including applicable valuation criteria. Cover pool eligible mortgages are generally first-lien mortgages only. In public sector loans, restrictions can apply to the type of public sector borrower, for instance whether or not the borrower has the right to levy discretionary taxes or fees. As a result of cover pool eligibility criteria, some retail mortgage markets are segmented. Banks would offer first-lien mortgage loans up to the

³This ratio, usually abbreviated as LTV, is the ratio of loan value to the most recent assessed property value. Property values are assessed by valuation experts using statutory valuation approaches. Some jurisdictions use indexation to regularly adjust property valuations on existing loans without a full assessment.

statutory LTV limit at one set of terms that reflect the covered bond funding costs, while the share of a mortgage loan that exceeds this LTV limit would be offered as a more expensive loan secured either by the same mortgage or an additional second lien mortgage⁴.

It is not immediately good or bad to have wider or narrower ranges of eligible collateral. A narrow eligible collateral range will lower the default risk in the cover pool but tend to result in higher asset concentration and lower net interest margin for the pool.

Unfortunately, national legislators have seen fit to create a proliferation of covered bond frameworks across even a comparatively harmonised financial landscape like the euro area. There are explanations for having different national frameworks, namely the interaction between insolvency law and covered bonds on one hand, and the long-term nature of cover pool assets on the other. German retail mortgages, for instance, tend to have contractual maturities of 30 years and are structured to match German cover pool criteria⁵. Harmonising frameworks across multiple countries would therefore require a wholesale switch in retail product designs. Such explanations are somewhat insufficient, however, for explaining multiple competing legislative frameworks in the same country.

25.2 DANISH COVERED BONDS

The Danish covered bond market is one of the oldest statutory markets in the world. While it has seen its share of innovation, some features make it very different from other markets.

The core distinguishing feature of Danish covered bonds is the balance principle which means that the underlying mortgages are closely related to the bonds with which they are being refinanced. A Danish borrower can therefore derive the likely terms of a new mortgage by observing the market prices of covered bonds⁶. In other markets mortgages of different types are bundled into the same cover pool and refinanced by portfolios of covered bonds so that this linkage in economic terms does not exist.

On the basis of the balance principle, Danish borrowers have an additional way of refinancing a mortgage through a process known as delivery. A borrower can purchase the specific bond that refinances her or his mortgage, and deliver this bond to the issuing bank in return for a full or partial cancellation of the mortgage loan. In addition,

⁴Jurisdictions differ in the precise legal definition of a mortgage and the existence or not of second-lien mortgages. In some cases a lender can take security over a property without a mortgage as such. Such differences are reflected in national covered bond frameworks.

⁵A German mortgage usually has a first-ranking portion covering up to 60% of the lending value eligible for Pfandbrief refinancing, and a second-ranking portion for amounts about this threshold which is refinanced in other ways, with a higher cost attached.

⁶One of the results of this close match is that Danish regulators were among the first to push for high price transparency for bonds which, as in other markets, trade largely over the counter (OTC). High transparency in this context means fast disclosure of a large subset of all trades conducted.

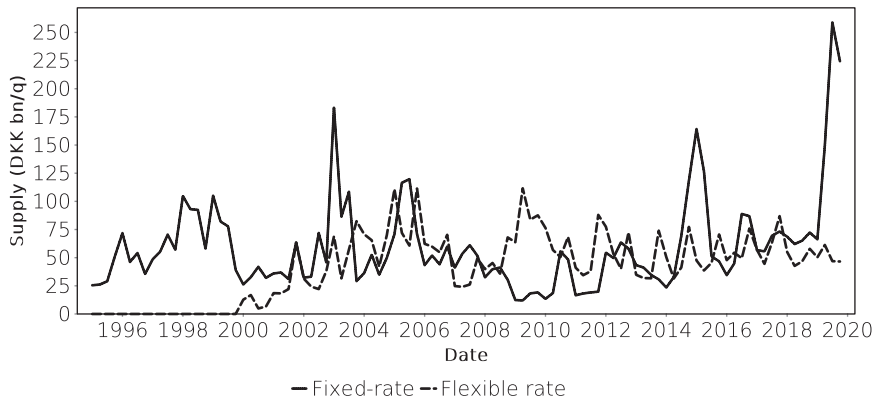


FIGURE 25.4 Quarterly gross Danish covered bond issuance volumes for fixed-rate and variable rate mortgages. Note that the shorter maturity of variable-rate refinancing bonds requires more frequent refunding.

Source: Finans Danmark Udlånsstatistikken.

borrowers can prepay the loans early, as is the case in US mortgages. In this case, the issuing bank will partially call the covered bond that refinances the mortgage that has been prepaid. These two prepayment mechanisms have opposing effects on a mortgage loan if done with the aim of refinancing the mortgage at current conditions with a new loan. Delivery does not change the present value of the mortgage, but when interest rates are higher than when the loan was originated, the mortgage can be restructured to a lower principal amount but a higher coupon. The main reason to conduct such a restructure is that it increases the economic value of the prepayment option. If interest rates fall again, the option to refinance the mortgage through prepayment, i.e., restructure the mortgage with the same principal but a lower coupon, usually has a positive value (cf. Section 27.1 on mortgage repayments on page 290).

The standard Danish mortgage contract is a 30Y annuity but this structure has been modified over the years. It is now common to have amortisation holidays within the contract, i.e., phases where the borrower makes only interest payments, to allow for lower income periods. The total length of such holidays is limited. Because long-dated annuities mean that borrowing rates are determined by the long end of the yield curve, a competing product has existed since 2000 that allows the borrower to save on the term risk premium at the expense of higher interest rate volatility. These contracts, known as Flexlån, have variable rate coupons with fixing periods of up to 10 years. The balance principle is implemented for such loans by re-fixing the coupon rate through auctions of the fixed-rate bonds that refund such mortgages. The relative share of fixed and variable-rate mortgages fluctuates over time in response to changing interest rates (see Figure 25.4).

Interestingly, although euro-denominated mortgages exist in Denmark and the Danish Krone is being kept in a very tight relationship with the euro through Denmark's membership in ERM II, the take-up of such euro-denominated mortgages is comparatively low.

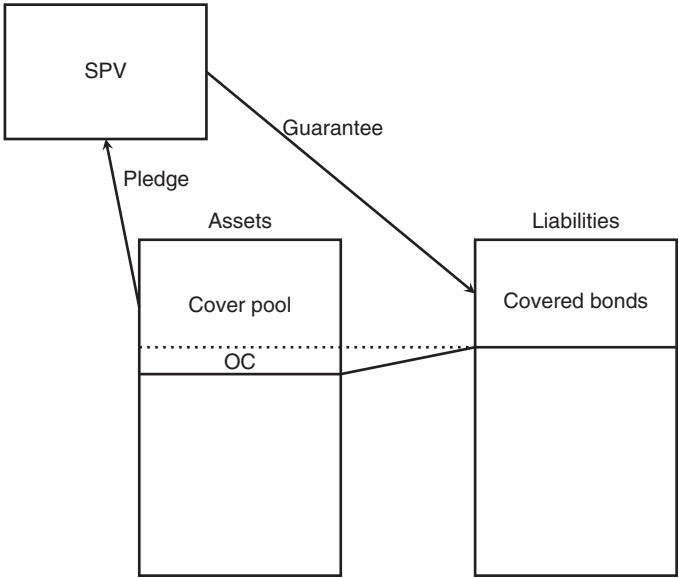


FIGURE 25.5 General structure of a structured covered bond issuer balance sheet and the guarantor SPV.

25.3 STRUCTURED COVERED BONDS

As mentioned above, structured covered bonds are private law arrangements that look to achieve the economic features of a covered bond. They are useful when a covered bond framework does not exist in the jurisdiction of a potential issuer, when the framework that does exist is unsuitable, or when there are specific circumstances that prevent an issuer to use an existing framework. An example of an unsuitable framework would be the absence of a carve-out of covered bond cover pools from standard insolvency procedures. Without this carve-out, covered bond investors would immediately join the general insolvency procedure and face an acceleration of their claims. An example of specific circumstances would be non-subordination language in an issuer’s senior unsecured bond documentation that prevent pledging assets to the cover pool.

The standard design pattern of a structured covered bond is a senior unsecured bank bond that is guaranteed by a special-purpose vehicle holding cover assets. Possible variations are in the funding of the assets on the SPV balance sheet, which is commonly done through an intercompany loan. In essence, the bank sets up the SPV, loans its funds and the SPV uses these funds to purchase the cover assets from the bank. The bank funds the loan through the issuance of a structured covered bond which is guaranteed by the SPV. Should the bank default on the bond, the SPV can use or liquidate the cover assets to fund payments under the guarantee it has extended to the holders of the structured covered bond. The terms of the intercompany loan are designed such that, while the banks, other creditors have a claim to repayment of the loan after insolvency of the bank these claims rank behind the guarantee. In this way, the residual

value of the cover pool (the economic equivalent of OC) falls to the unsecured creditors of the bank, just as in any other covered bond structure. This distribution only takes place after the maturity of the covered bond, in line with most statutory covered bond frameworks.

In some jurisdictions, the asset transfer to the guarantor SPV is only perfected at the time of default of the issuer. In this case, the intercompany loan does not occur at the time of bond issuance, and the delay in the asset transfer can reduce tax or accounting issues that would be faced by the issuer if the asset transfer happened at the time of bond issuance.

An important use of structured covered bonds is the ‘recycling’ of suitable loan pools that were not originated with covered bond issuance in mind. Statutory covered bond frameworks specify features for eligible loans while structured covered bonds can use loans that vary in some respects, such as geographical location, type of collateral, or loan-to-value ratios. Because the assets of structured covered bonds are transferred to the SPV rather than being pledged, structured covered bonds are also a possible avenue for issuers with negative pledge clauses on existing senior unsecured debt to issue covered bonds.

25.4 COVERED BOND CREDIT RISK ANALYSIS

The credit risk of a covered bond is driven by three factors in addition to its maturity: The strength of the issuing bank, the quality of the cover pool assets, and the amount of OC. Simply speaking, a perfectly safe bank will never default, so the existence or quality of the cover assets is irrelevant⁷. If the bank is not perfectly safe, then the credit quality is determined by the likelihood of cover assets sufficing to generate the cash payments due under the bonds, even after the issuer is no longer in a position to replace deteriorated cover assets. Again simply speaking, the faster the cover assets deteriorate, the more OC is required to preserve sufficient cover assets until the maturity of the covered bond. Also, with equal cover asset quality, the earlier the issuing bank is expected to default, the more OC will be used up before the covered bonds mature. Figure 25.6 illustrates this logic.

A crucial feature of a covered bond is that, in contrast to an ABS, the collateral pool is constantly replenished through asset substitution by the issuing bank. This replenishment stops when the issuing bank defaults. At that point, the value of the cash flows from the collateral pool declines through credit deterioration, similar to the rating evolution shown in Figure 24.2 on page 256. The speed at which this deterioration happens is a function of the quality of the credit pool. What matters for the credit quality of the covered bonds is whether this deterioration process erodes the collateral value below the payment obligation of the bond before the bond matures. Aside from the quality of the cover pool, this is determined by the expected time between the default and the

⁷In the Japanese market, where the large banks are traditionally seen as very safe (or at least too big to fail), covered bond development was held back by very tight senior unsecured pricing. The market simply had not developed a demand for cover assets.

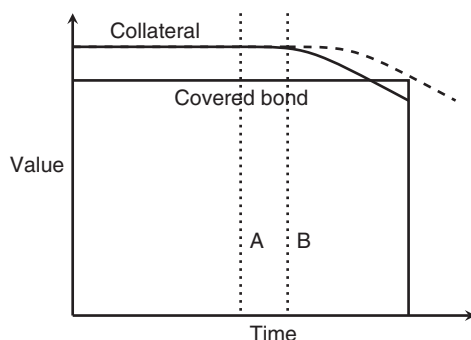


FIGURE 25.6 Illustration of the three drivers of covered bond default risk for two default timing scenarios of the issuing bank for a single outstanding covered bond. In the case of an earlier default (A), the excess collateral is exhausted before the maturity of the bond and bond holders face losses. In the case of a later default (B), the excess collateral declines at the same speed after default but is still sufficient for repayment of the bond.

maturity of the bond, which is a function of the stand-alone credit quality of the issuing bank, and the amount of excess collateral. Therefore, simply speaking, a riskier collateral pool, longer-dated covered bonds, and a lower stand-alone credit quality of the bank require more excess collateral to achieve the same credit quality for the covered bond.

It should be borne in mind that the choice of cover assets is a business decision of the issuing bank. A given bank may opt to finance riskier loans, and so deliberately include lower quality assets in the cover pool, than other banks. As long as the amount of excess collateral appropriately reflects this choice, this difference in cover assets need not affect the credit quality of the covered bond. This means also that more excess collateral is not necessarily a sign of a weaker quality of the bank, although it might be.

The issue of asset encumbrance also affects the structuring choices of the bank. Pledging more excess collateral potentially increases the riskiness of lower-ranked debt like senior unsecured bonds. As long as markets price this risk, issuing more senior unsecured debt to fund the excess collateral becomes more expensive as a result of this encumbrance. This affects the economics of the covered bond issuance, and in effect provides a cap on credit enhancement of covered bonds through the use of excess collateral. Using low-quality collateral for covered bonds is uneconomical due to the cost of funding the excess collateral with senior unsecured debt and equity.